Meta-Learning for Quantitative Trading

Applying Meta-SGD with SAM + L1 Sparsity + Online Adaptation to the Numin2 Nifty 50 Dataset

+43.20	0.493	1,489
TOTAL PROFIT	SHARPE RATIO	OOS DAYS

META & TRANSFER LEARNING · NUMIN2 ASSIGNMENT

Agenda

01	Problem & Dataset	What are we solving?
02	Data Pipeline & Split	How data flows in
03	Baseline Approach	Starting point
04	Our Architecture	Meta-SGD + SAM
05	Key Experiments & Ablations	What we tried
06	Final Results & Analysis	What worked
07	Conclusion	Takeaways

Problem Statement

The Task

For each of the **50 Nifty stocks**, predict a daily signal:

- +1 = Buy the stock
- 0 = Hold / Stay in cash
- −1 = Sell the stock short

Why Meta-Learning?

- Markets exhibit **non-stationary regimes** - bull, bear, sideways
- A static model trained on 2010 data *fails* in 2023
- Meta-learning lets the model **adapt to new regimes** at inference time using just 1 gradient step

Available Data

OHLCV + Daily Returns for 50 Nifty stocks

~10,000 trading days · 20 years · 6 features per stock

Data Pipeline & Temporal Split

RAW OHLCV
Open, High, Low, Close, Volume

→

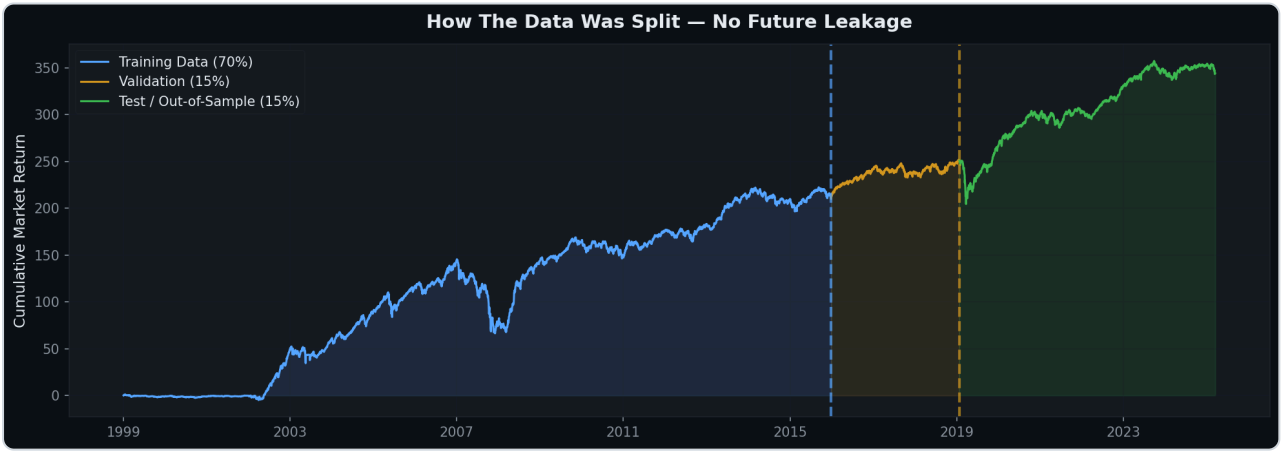
FEATURE BLOCK
GK Volatility, Rel Volume, Z-Score, 60d Momentum

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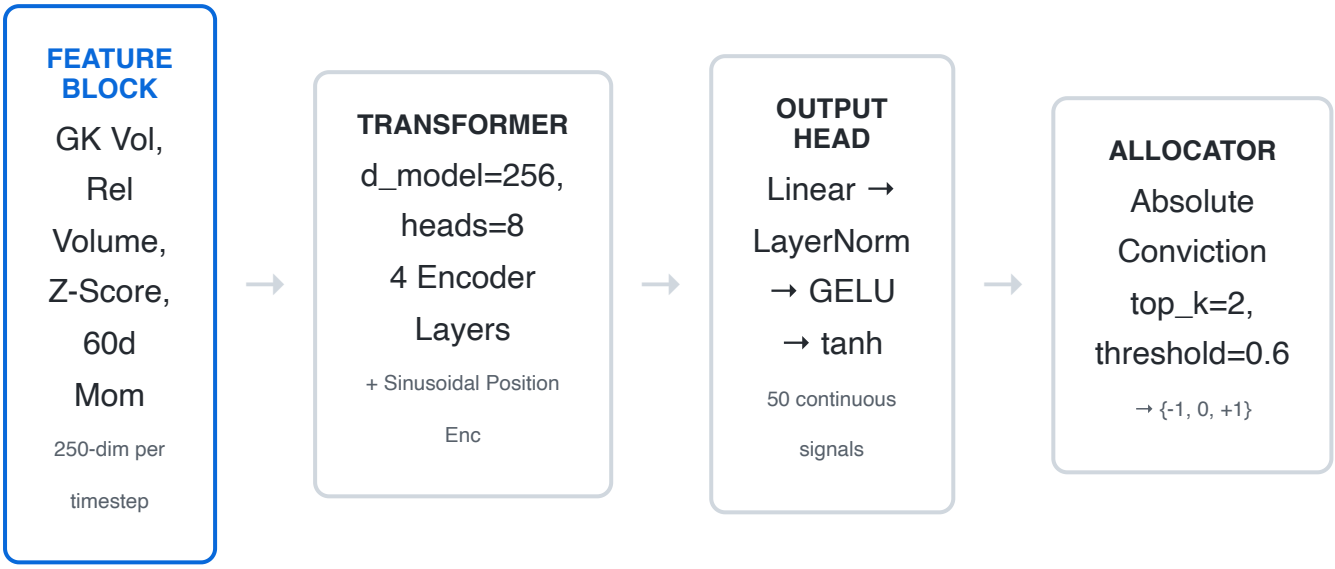
TEMPORAL SPLIT
70% Train
15% Val · 15% Test

→

DATALOADER
252-day sliding windows
Batch size: 128



Our Architecture — Meta-SGD Transformer



Phase 1: Meta-Training (Formulation)

- **Inner Loop:** Adaptation on batch *i* via learned inner LR α:

$$\theta' = \theta - \alpha \odot \nabla L_{inner}$$

- **Outer Loop:** Evaluate weights θ' on batch *i+1* via **Turnover-Aware Reward:**

$$Net_Reward = (p_t \cdot y_t) - c \times |p_t - p_{t-1}|$$

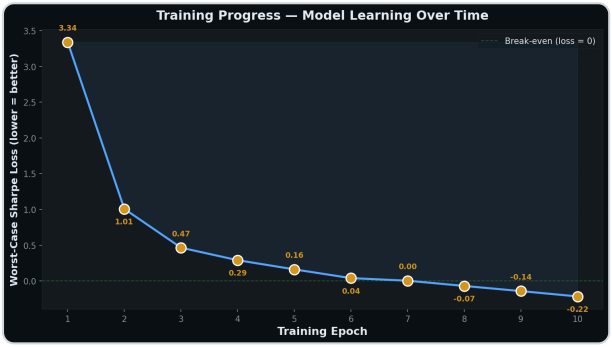
- **SAM:** Sharpness-Aware Minimization (ρ=0.05) finds *flat minima*.
- **L1 Sparsity:** Penalty forces 0.0 outputs for non-confident plays.

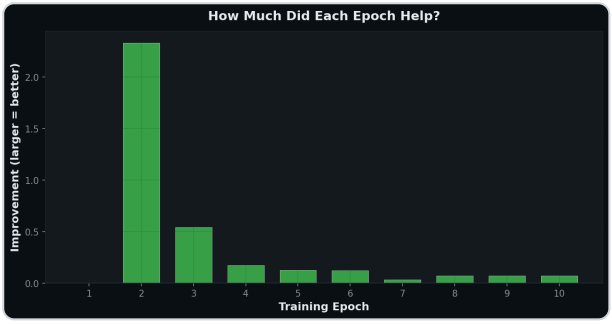
Phase 2: Online Adaptation (Inference)

- Before predicting day *t*, the model **adapts to day t–1**
- Uses the *learned inner learning rates* (per-parameter) for a single gradient step
- Adapted weights are used via `functional_call()` — global weights untouched
- The model **tracks the current market regime in real-time**

Online Adaptation was the **single most impactful technique** — it flipped the OHLCV model from -1.64 to +25.84 profit.

Training Progress





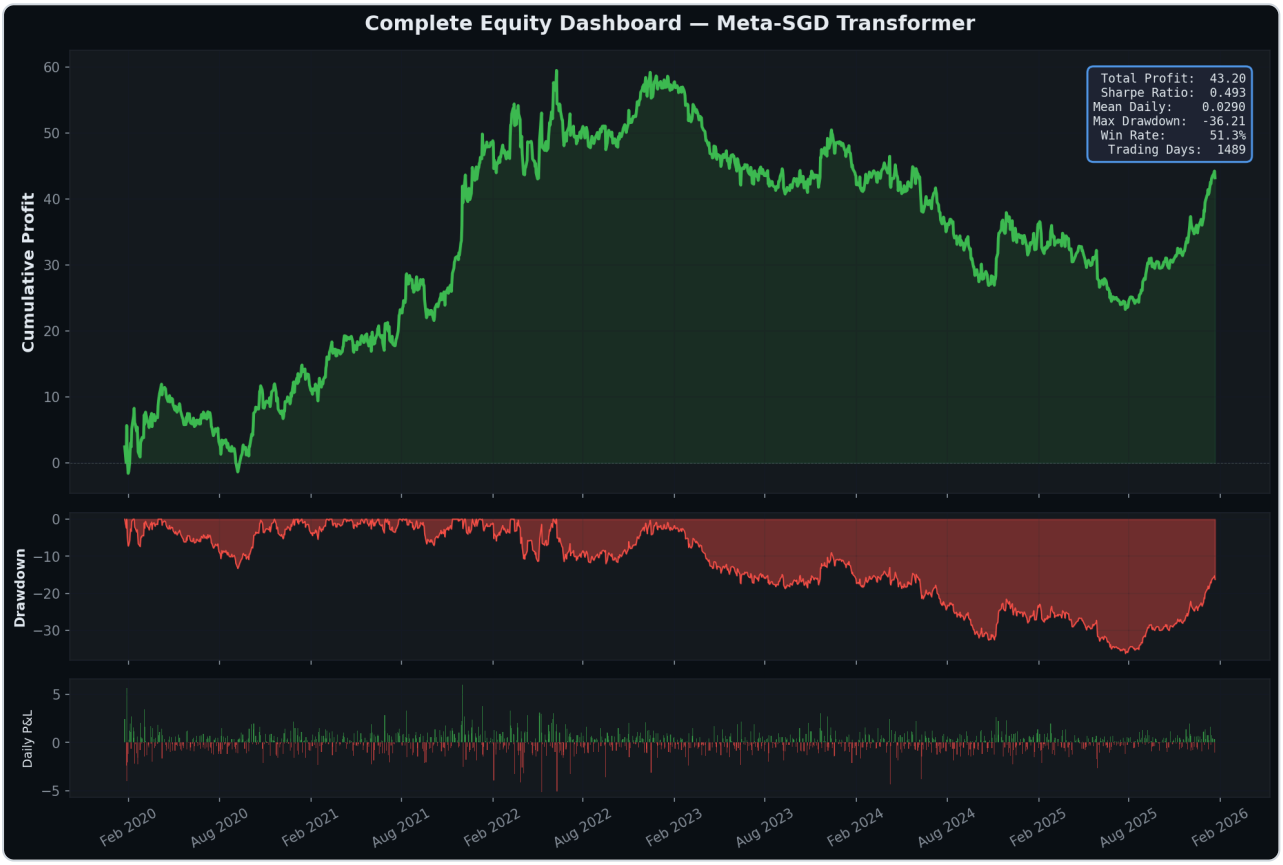
What This Shows

- Worst-case Sharpe loss drops from **3.34** → **-0.22**
- Negative loss = the model generates *positive* Sharpe even in worst-case SAM perturbations
- Biggest jump in **Epoch 2** (Δ=2.33), steady refinement after
- No overfitting — loss monotonically decreases

Key Experiments & Ablation Study

#	Configuration	Allocator	Top-K	Inertia	Total Profit	Sharpe
1	Final Model (No OHLCV)	Absolute	2	0.4	+20.58	0.227
2	Full OHLCV + Meta-SGD + SAM	Strict Discrete	4	—	-1.64	-0.024
3	Online Adaptation + Lazy StatArb	Lazy StatArb	2	1.5	+25.84	0.34
FINAL	**Meta-SGD + SAM + L1 + Online Adapt**	**Absolute**	**2**	**0.4**	+43.20	0.493

Final Model — Equity Curve



+43.20	0.493	51.3%	-36.2	1,489
TOTAL PROFIT	SHARPE RATIO	WIN RATE	MAX DRAWDOWN	OOS DAYS

Strategy vs Market Benchmark



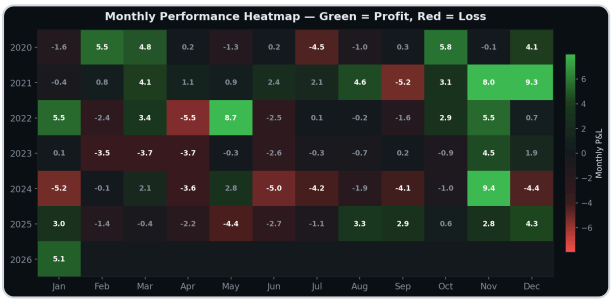
Why It Outperforms

- The model learned to **sit in cash** during uncertain periods
- Active positions only when **conviction > 0.6**
- The inertia bonus prevents **unnecessary rotation**

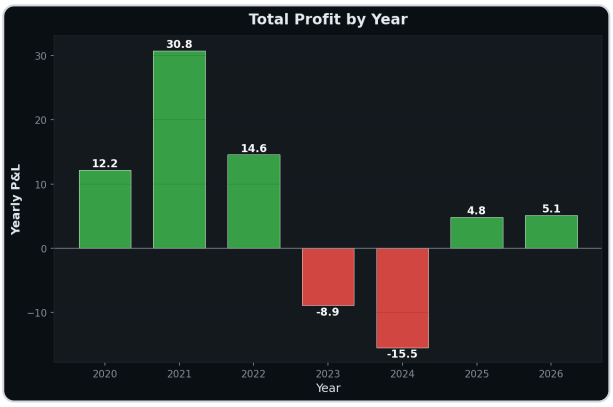
The Nifty 50 Comparison

- Equal-weight Nifty 50 = passive buy-all exposure
- Our strategy is **selective** — only 2 stocks max at a time
- Lower drawdowns during market crashes

Performance Breakdown



Monthly P&L Heatmap — Green = profit, Red = loss

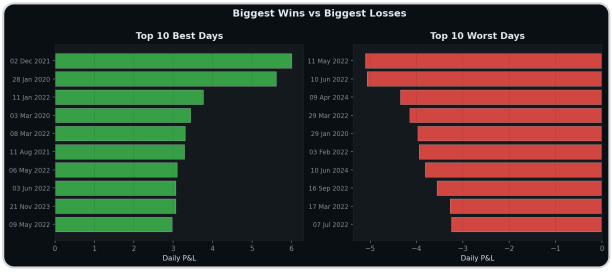


Total profit by year

Risk & Consistency Analysis



Rolling Sharpe Ratio over time



Top 10 best vs worst individual days

Key Insights & Novelty

1. Sparsity is Alpha

Adding an L1 penalty to the meta-loss forces the model to output **0.0 (cash)** by default. It trades only when confident.

Impact: -17 → +42 profit

2. Online Adaptation

Before predicting day *t*, the model adapts to day *t−1* via learned per-parameter inner LRs. This is the key meta-learning contribution.

Impact: -1.64 → +25.84 profit

3. SAM for Flat Minima

Sharpness-Aware Minimization ensures broad, flat loss basins that survive market regime changes without catastrophic forgetting.

Prevents regime-crash drawdowns

4. The Allocator Matters

The same neural backbone produces wildly different profits depending on the discrete allocator. The **Absolute Conviction Allocator** (threshold=0.6, top_k=2, inertia=0.4) was the winning formula — defaulting to cash unless overwhelmingly confident.

5. Feature Engineering is Essential

The final model trained without OHLCV data achieved a Sharpe of only **0.227**. Incorporating full OHLCV volume structures with cross-sectional Z-scoring was necessary to reach **0.493**.

Conclusion

Final Performance

+43.20
TOTAL PROFIT

0.493
SHARPE

Competitive with classic momentum baseline (Sharpe ~0.5) despite 10bps fee.

Meta-Learning Value

- **Meta-SGD:** Learned per-parameter inner LRs — no manual tuning of adaptation rates
- **Online Adaptation:** Real-time regime tracking at inference — the model is never stale
- **SAM:** Robust flat-minima → survives distribution shifts

Lessons Learned

- Transaction costs are a **first-order problem**, not an afterthought
- The allocator design (how to map signals → positions) is as critical as the neural network
- Sparsity + inertia = the key to surviving friction-heavy environments
- Deep feature engineering (Z-scoring, relative vol) is mandatory to beat market noise

Future Work

- Integrate attention-based allocator to learn position sizing end-to-end
- Multi-horizon Meta-SGD (adapt across both short-term and long-term regimes)
- Apply to other universes (S&P 500, Crypto)

Thank You

Questions & Discussion

+43.20	0.493	51.3%	9 Expts
TOTAL PROFIT	SHARPE RATIO	WIN RATE	ABLATED

META-SGD · SAM · ONLINE ADAPTATION · NUMIN2 API